谢应明（Ying-Ming Xie）

上海交通大学物理与天文学院，上海 *200240*，中国
Email：yingming.xie@sjtu.edu.cn
Tel：(+86) 15178790277
Google Scholar，arXiv，个人主页

工作经历

06/2026 ～今	副教授 上海交通大学物理与天文学院凝聚态物理研究所
07/2023 ～06/2026	理研（RIKEN）基础特别研究员 SPDR Fellow 导师：Naoto Nagaosa
05/2022 ～06/2023	香港研资局（RGC）博士后奖学金获得者 香港科技大学（HKUST）
10/2021 ～04/2022	香港科技大学博士后研究员（导师：Law Kam Tuen）

相关研究经历

06/2023	访问博士后研究员，麻省理工学院 MIT 导师：Patrick A. Lee
10/2022 ～01/2023	访问博士后研究员，加州理工学院 Caltech 导师：Jason Alicea

教育背景

2016.09 ～2021.10	香港科技大学物理学博士（PhD），导师：Law Kam Tuen
2016.02 ～2016.06	香港科技大学，本科科研实习
2012.09 ～2016.07	北京师范大学，物理学学士（B.Sc.）

研究兴趣

研究方向包括：非常规超导与磁性、拓扑超导与 Majorana 费米子、约瑟夫森结、拓扑材料、二维/莫尔材料、密度波、非线性光学、强关联体系、量子信息在凝聚态物理中的交叉应用等。

学术服务

审稿人（次数）：Nature（2）、Phys. Rev. Lett.（4）、Nat. Commun.（2）、Phys. Rev. B（7）、Phys. Rev. Applied（1）、2D Materials（1）。

获奖与荣誉

- 日本理化学研究所基础科学特别研究员（SPDR Fellowship），RIKEN（2023）
- 香港 RGC 博士后奖学金（2022）
- 香港科技大学理学院博士研究卓越奖（2021）
- George K. Lee Foundation Scholarship（2019）
- 北京师范大学优秀本科生（2016）
- 国家奖学金（2013，2014）

学术报告

03/2026	邀请 Colloquium 报告：When topology meets correlations: emergent quantum matter in two dimensions 上海交通大学物理与天文学院 中国上海
08/2025	邀请报告：Photon-drag photovoltaic effects and quantum geometric nature 第 30 届国际低温物理大会（LT30） 西班牙 Bilbao
05/2025	邀请报告：Unconventional Josephson effects and FFLO in Moiré crystals 中国科学技术大学国际功能材料量子设计中心 中国合肥
03/2024	报告：Gate-Defined Topological Josephson Junctions in Bernal Bilayer Graphene 美国物理学年会（APS March Meeting），美国明尼阿波利斯

08/2023 邀请报告：Orbital Fulde-Ferrell Pairing State in Moiré Ising Superconductors
理研（RIKEN）CEMS AB Meeting，日本

05/2023 邀请报告：Theory of Interaction-Driven Quantum Anomalous Hall States in MoTe2/WSe2 Hetero-Bilayers
Gordon Research Seminar，美国 Ventura

03/2023 报告：Pauli Limit Violation, Finite-Momentum Pairing, and Superconducting Diode Effect in Moiré Ising Superconductors
美国物理学年会（APS March Meeting），美国拉斯维加斯

11/2022 邀请报告：Valley-Polarized Quantum Anomalous Hall States and Unconventional Josephson Effects in Moiré Materials
Caltech CMT Seminar，美国加州理工学院

03/2021 报告：Spin-Orbit-Parity-Coupled Superconductivity in Topological Monolayer WTe2
美国物理学年会（APS March Meeting）（线上）

03/2019 报告：Enhancement of In-Plane H_{c2} in Monolayer Centrosymmetric Superconductor 1T'-WTe2 via Strong Spin-Orbit Coupling
美国物理学年会（APS March Meeting），美国波士顿

03/2018 报告：Quasi-One-Dimensional Quantum Anomalous Hall Systems as New Platforms for Scalable Topological Quantum Computation
美国物理学年会（APS March Meeting），美国洛杉矶

论文发表列表（更新至 2026 年 6 月）

1. Dong Li, Yuki M. Itahashi, **Ying-Ming Xie**, Menghu Zhou, Yukako Fujishiro, Kunjie Zheng, Yao Guang, Max T. Birch, Ilya Belopolski, Max Hirschberger, Takahiro Morimoto, Xiao-Xiao Zhang, Zhi-An Ren, Naoto Nagaosa, and Yoshihiro Iwasa, Surface-switchable nonreciprocity protected by Fermi arcs in Weyl semimetal TaAs, arXiv:2606.17395 (2026).

2. Zi-Ting Sun, **Ying-Ming Xie**, and Naoto Nagaosa, Quantum Information Geometry of Multicomponent Superconducting Fluctuation Transport, arXiv:2606.15928 (2026).

3. Luca Maranzana, Koki Shinada, **Ying-Ming Xie**, Sergey Artyukhin, and Naoto Nagaosa, Semi-classical Wave-Packet Dynamics in Phase-Space Geometry: Quantum Metric Effects, arXiv:2603.21262 (2026).

4. **Ying-Ming Xie** and Naoto Nagaosa, Theory of Little-Parks oscillations by vortices in two-dimensional superconductors, arXiv:2601.23050 (2026).

5. Grant Z. X. Yang, Zi-Ting Sun, **Ying-Ming Xie**, and K. T. Law, Topological altermagnetic Josephson junctions, *npj Quantum Materials* **11**, 45 (2026).

6. **Ying-Ming Xie**, Naoto Nagaosa, Probing loop currents and collective modes of charge density waves in Kagome materials with NV centers, *npj Quantum Materials* **10**, 64 (2025).

7. Jin-Xin Hu, Mengli Hu, **Ying-Ming Xie**, and K. T. Law, Unconventional Josephson effects in PT-symmetric antiferromagnetic bilayers, arXiv:2509.08262 (2025).

8. Enze Zhang, Zi-Ting Sun, Zehao Jia, Jinshan Yang, Jingyi Yan, Linfeng Ai, **Ying-Ming Xie**, Yuda Zhang, Xue-Jian Gao, Xian Xu, Shanshan Liu, Qiang Ma, Chaowei Hu, Xufeng Kou, Jin Zou, Ni Ni, Kam Tuen Law, Shaoming Dong, Faxian Xiu, Observation of edge supercurrent in topological antiferromagnet MnBi2Te4-based Josephson junctions, Science Advances 11, eads8730 (2025).

9. Jin-Xin Hu[#], Akito Daido, Zi-Ting Sun, **Ying-Ming Xie**[#], and K.T. Law[#], Layer Pseudospin Superconductivity in Twisted MoTe2, arXiv:2506.12767 (2025).[# Corresponding author]

10. Pseudo-Ising superconductivity induced by p-wave magnetism, Zi-Ting Sun, Xilin Feng, **Ying-Ming Xie**, Benjamin T. Zhou, Jin-Xin Hu, K. T. Law, *Phys. Rev. B* **112**, 214504 (2025).

11. **Ying-Ming Xie**, Naoto Nagaosa, Quasi-one-dimensional Supersolids in Luther-Emery Liquids, *Phys. Rev. Research* **7**, 023302 (2025).

12. **Ying-Ming Xie**, Naoto Nagaosa, Photon Drag Photovoltaic Effects and Quantum Geometric Nature, *PNAS* **122**, e2424294122 (2025).

13. **Ying-Ming Xie**, Hiroki Isobe, Naoto Nagaosa, Nonreciprocal nonlinear responses in moving charge density waves, *npj Quantum Materials* **9**, 82 (2024).

14. **Ying-Ming Xie**, Naoto Nagaosa, Phase shifts, band geometry and responses in triple-Q charge and spin density waves, *Phys. Rev. B* 110, L241108 (2024).

15. **Ying-Ming Xie**, Yizhou Liu, and Naoto Nagaosa, Sliding Dynamics of Current-Driven Skyrmion Crystal and Helix in Chiral Magnets, *Phys. Rev. Lett.* **133**, 096702 (2024).

16. Zi-Ting Sun, Jin-Xin Hu, **Ying-Ming Xie** [#] , and K. T. Law [#], Anomalous $h/2e$ Periodicity and Majorana Zero Modes in Chiral Josephson Junctions, *Phys. Rev. Lett.* 133, 056601 (2024). [# Corresponding author]

17. **Ying-Ming Xie**, Cheng-Ping Zhang, Kam Tuen Law, Topological $p_x + ip_y$ inter-valley coherent state in Moiré MoTe2/WSe2 heterobilayers, *Phys. Rev. B* **110**, 045115 (2024).

18. Christian Tzschaschel, Jian-Xiang Qiu, Xue-Jian Gao, Hou-Chen Li, Chunyu Guo, Hung-Yu Yang, Cheng-Ping Zhang, **Ying-Ming Xie**, Yu-Fei Liu, Anyuan Gao, Damien Bérubé, Thao Dinh, Sheng-Chin Ho, Yuqiang Fang, Fuqiang Huang, Johanna Nordlander, Qiong Ma, Fazel Tafti, Philip JW Moll, Kam Tuen Law, Su-Yang Xu, *Nat. Commun.*, 3017 (2024).

19. Omargeldi Atanov, Wai Ting Tai, **Ying-Ming Xie**, Yat Hei Ng, Molly A Hammond, Tin Seng Manfred Ho, Tsin Hei Koo, Hui Li, Sui Lun Ho, Jian Lyu, Sukong Chong, Peng Zhang, Lixuan Tai, Jiannong Wang, Kam Tuen Law, Kang L Wang, Rolf Lortz, Proximity-induced quasi-one-dimensional superconducting quantum anomalous Hall state, *Cell Reports Physical Science* 5.1 (2024).

20. **Ying-Ming Xie**, Étienne Lantagne-Hurtubise, Andrea F. Young, Stevan Nadj-Perge, and Jason Alicea, Gate-Defined Topological Josephson Junctions in Bernal Bilayer Graphene, *Phys. Rev. Lett.* **131**, 146601 (2023).

21. Jin-Xin Hu, Zi-Ting Sun, **Ying-Ming Xie**#, and K. T. Law#, Josephson Diode Effect Induced by Valley Polarization in Twisted Bilayer Graphene, *Phys. Rev. Lett.* **130**, 266003 (2023). [# Corresponding author]

22. **Ying-Ming Xie**, K. T. Law, Orbital Fulde-Ferrell pairing state in moiré Ising superconductors, *Phys. Rev. Lett.* **131**, 016001 (2023).

23. J. Díez-Mérida, A. Díez-Carlón, S. Y. Yang, **Y.-M. Xie**, X.-J. Gao, K. Watanabe, T. Taniguchi, X. Lu, K. T. Law and Dmitri K. Efetov. Magnetic Josephson Junctions and Superconducting Diodes in Magic Angle Twisted Bilayer Graphene. *Nat. Commun.* **14**, 2396 (2023).

24. **Ying-Ming Xie**, Dmitri K. Efetov, K. T. Law., φ_0–Josephson junction in twisted bilayer graphene induced by a valley-polarized state, *Phys. Rev. Research* **5**, 023029 (2023).

25. Cheng-Ping Zhang, Xue-Jian Gao, **Ying-Ming Xie**, Hoi Chun Po and K. T. Law. Higher-Order Nonlinear Anomalous Hall Effects Induced by Berry Curvature Multipoles. *Phys. Rev. B* **107**, 115142 (2023).

26. Jin-Xin Hu, **Ying-Ming Xie**, and K. T. Law, Berry curvature, spin Hall effect, and nonlinear optical response in moiré transition metal dichalcogenide heterobilayers, *Phys. Rev. B* **107**, 075424 (2023).

27. Enze Zhang*, **Ying-Ming Xie***, Yuqiang Fang*, Jinglei Zhang*, Xian Xu, Yi-Chao Zou, Pengliang Leng, Xue-Jian Gao, Yong Zhang, Linfeng Ai, Yuda Zhang, Zehao Jia, Shanshan Liu, Wei Zhao, Sarah J. Haigh, Xufeng Kou,Jinshan Yang, Fuqiang Huang, K. T. Law, Faxian Xiu, Shaoming Dong. Spin-Orbit-Parity Coupled Superconductivity in Atomically thin 2M-WS2. *Nat. Phys.* **19**, 106-113 (2023). [$*$ equally contribution]

28. Jin-Xin Hu, Cheng-Ping Zhang, **Ying-Ming Xie**, K. T. Law. Nonlinear Hall Effects in Strained Twisted Bilayer WSe2, *Communications Physics* **5**, 255 (2022).

29. Cheng-Ping Zhang, Jiewen Xiao, Benjamin T. Zhou, Jin-Xin Hu, **Ying-Ming Xie**, Binghai Yan, and K. T. Law. Giant nonlinear Hall effect in strained twisted bilayer graphene. *Phys. Rev. B*: **106**(4), L041111 (2022).

30. **Ying-Ming Xie**, Cheng-Ping Zhang, Jin-Xin Hu, Kin Fai Mak, and K. T. Law, Valley-Polarized Quantum Anomalous Hall State in Moiré MoTe2/WSe2 Heterobilayers. *Phys. Rev. Lett.* **128**, 026402 (2022).

31. **Ying-Ming Xie**, Xue-Jian Gao, Tai-Kai Ng, K. T. Law. Creating majorana zero modes in quantum anomalous hall insulator/superconductor heterostructures. US Patent App. 17/672,434 (2022).

32. **Ying-Ming Xie**, K. T. Law, and Patrick A. Lee. Topological superconductivity in EuS/Au/superconductor heterostructures, *Phys. Rev. Research* **3**, 043086 (2021).

33. **Ying-Ming Xie**, Xue-Jian Gao, Xiao Yan Xu, Cheng-Ping Zhang, Jin-Xin Hu, and K. T. Law. Kramers nodal line metals, *Nat. Commun.* **12**, 3064 (2021).

34. **Ying-Ming Xie**, Benjamin T Zhou, K. T. Law. Spin-orbit-parity coupled superconductivity in topological monolayer WTe2. *Phys. Rev. Lett.* **125**, 107001 (2020).

35. Yong-Bin Choi*, **Ying-Ming Xie***, Chui-Zhen Chen, Jinho Park, Su-Beom Song, Jiho Yoon, Bum Joon Kim, Takashi Taniguchi, Kenji Watanabe, Jonghwan Kim, Kin Chung Fong, Mazhar N Ali, Kam Tuen Law, Gil-Ho Lee. Evidence of higher-order topology in multilayer WTe 2 from Josephson coupling through anisotropic hinge states. *Nature Materials* **19**, 974–979 (2020). [$*$ equally contribution].

36. Manna, Sujit, Peng Wei, **Ying-Ming Xie**, Kam Tuen Law, Patrick Lee, and Jagadeesh Moodera. Signature of a pair of Majorana zero modes in superconducting gold surface states, *Proceedings of the National Academy of Sciences* **117**, 8775 (2020).

37. Junying Shen, Jian Lyu, Jason Z Gao, **Ying-Ming Xie**, Chui-Zhen Chen, Chang-woo Cho, Omargeldi Atanov, Zhijie Chen, Kai Liu, Yajian J Hu, King Yau Yip, Swee K Goh, Qing Lin He, Lei Pan, Kang L Wang, Kam Tuen Law, and Rolf Lortz. Spectroscopic fingerprint of chiral Majorana modes at the edge of a quantum anomalous Hall insulator/superconductor heterostructure, *Proceedings of the National Academy of Sciences* **117**, 238 (2020).

38. **Ying-Ming Xie**, Benjamin T Zhou, T. K. Ng, and K. T. Law. Strongly enlarged topological regime and enhanced superconducting gap in nanowires coupled to Ising superconductors, *Physical Review Research* **2**, 013026 (2020).

39. Chui-Zhen Chen, **Ying-Ming Xie**, Jie Liu, Patrick A. Lee, and K. T. Law. Quasi-one-dimensional Quantum Anomalous Hall Systems as New Platforms for Scalable Topological Quantum Computation, *Phys. Rev. B* **97**, 104504 (2018).